Zenith: testing repairs to an adaptive investigator

Will Cook • 13 October 2026

Abstract. Can an investigator improve its reusable procedure by testing a diagnosis of its own work? Zenith implements a small loop: Lab chooses numerical investigations, Oracle reviews the recorded computation against later observations, and Evolve proposes an exact procedure replacement. Code reruns the whole investigation under unchanged and candidate procedures and returns the measured difference to the next review. The first historical episode completed this loop but did not improve forecasts. Its reviewer misread numerical evidence and repeated an unsupported diagnosis after an unchanged replay. We report this failure, the executable design and the comparisons needed to establish learning. A completed feedback loop is an experimental instrument, not evidence of improvement.

1. Start with a competent investigator. A repair system needs something worth repairing. We supply a working investigator, accurate numerical tools, sensible references and an earlier worked example before evaluating model-authored changes. These are engineering choices, not discoveries attributed to the learning loop.

Lab is one adaptive model using three actions. It can *inspect* observations, *compare* a registered numerical representation, or *commit* forecasts. Each answer is available before the next action. A question must say what its possible answers would change. Lab can stop early, retain a reference or investigate across baskets; there is no compulsory question for each target.

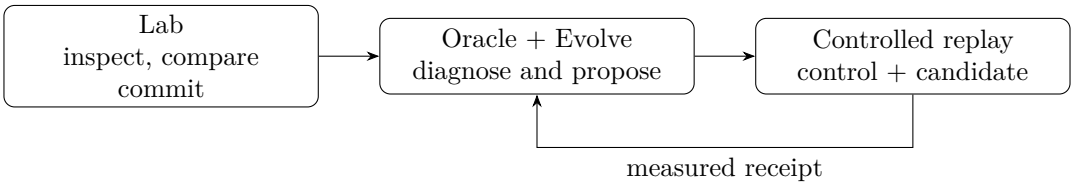
The task forecasts every basket's five-session return from the shared history of 147 eligible NEPSE instruments and 17 indices [1]. Six research baskets have membership fixed using earlier observations and fixed shares following equal initial-value allocation. For value V_b, the target is $r_{b,t} = V_b(t+5)/V_b(t) - 1$. These are constructed indices, not traded ETFs. Missing quotes retain their last mark and expose its age.

The tools preserve distinctions that matter: past returns versus future targets; member participation versus value contribution; predictive association versus accounting; and improvement over a weak comparator versus improvement over the selected reference. A distinction earns its place when it changes a useful question, decision or test. Requiring a long reasoning schema at every step is not itself a scientific contribution.

The numerical boundary. Code supplies ridge, zero-return and mature-training-mean references for each basket. Its advisory default minimises past validation error. An inspection returns constituent contributions and dated features. A comparison adds one summary—positive value weight, top-three absolute-contribution share or fresh quote weight—from any basket or the whole universe to the same ridge model. It keeps feature-count normalisation, regularisation, training rules and eight validation origins fixed, and reports paired results for all six targets.

The model chooses registered candidate IDs; code supplies their forecasts and target identities. Commitment requires all six targets. Invalid output stops the attempt without retry or a substitute forecast. Repeated searches over eight origins remain exploratory model selection. Price histories do not establish market causes.

2. One procedure, one measured repair. The native engine executes action/result pairs, with at most eight model turns and six comparisons. Early commitment makes the remaining slots inactive. The exported graph records the calls and supplied contexts that actually executed; its edges describe information flow, not economic causality. Public justifications and tool receipts are inspectable records, not access to the model's private internal reasoning.



Oracle diagnosis and Evolve's proposal share one model response. The reviewer receives the executed Lab graph, exact instructions and inputs, actions, numerical receipts and a separately labelled later outcome. A wrong forecast does not by itself locate an error. A proposed defect must cite an actual step, consider an alternative explanation and remain within available capabilities. No supported repair is a valid conclusion.

The editable object is one reusable investigation procedure of at most 4,096 characters. A candidate replaces it in full. The reviewer cannot silently alter the tools, model, targets or evaluator. Its proposal names an expected first action signature, an interpretation, a valid counterexample and a disconfirmer. The signature is a limited observable: an action occurring does not establish that the reasoning improved.

The coordinator freezes the procedure, snapshot, runtime, model and resource limits. It runs the complete Lab from entry with both the unchanged procedure and the exact proposed text. This tests changes to question selection as well as final choices. The receipt separates changed actions, numerical recipes, forecasts and losses. It retains failures. No candidate is automatically adopted.

The next actual reviewer request contains that exact receipt, verified in both the saved input and provider prompt. The reviewer must assess the tested proposal before revising, rejecting, retaining it for evaluation or making no change. Delivery is necessary for feedback use; it does not prove that the model understood or used it.

3. What would count as learning? Engineering fixes to tools or prompts first create a new baseline. Learning is then a model-authored procedure change evaluated within a frozen baseline. Keep these claims separate:

- **Execution:** the exact proposal ran and its measured effects reached the next review.
- **Behaviour:** the intended investigation or decision changed beyond ordinary repeated-run variation.
- **Predictive value:** the complete forecast vector improved under the declared loss and cost accounting.
- **Transfer:** a frozen reusable change helped on protected further cases, including counterexamples and regressions.

A same-case repair chosen after seeing the outcome supplies development evidence. One unchanged/candidate pair cannot establish a stochastic treatment effect. Numerical enumeration is an essential simpler comparator, with any larger search allowance disclosed. A multi-agent architecture must earn its extra cost against this adaptive controller using the same observations and tools.

4. First execution and its failure. On the exposed 2 February 2025 case, GPT-4.1 inspected the whole universe, compared B04's contribution concentration, and committed zero-return references for all six baskets. Zero had the lowest reported validation error among the available options.

The reviewer nevertheless claimed that other references had lower error. For example, it treated B03's mean-reference error 0.0006221 as lower than zero's 0.0005136. It proposed a procedure requiring minimum-validation-error selection. That diagnosis was false, and the repair was unnecessary on this case.

Condition	Mean squared return error	Observation
Original investigator	0.0006426687	Six zero-return forecasts
Unchanged whole-Lab replay	0.0006426687	Same forecasts
Proposed-procedure replay	0.0006426687	Same forecasts
Numerical enumeration	0.0006426687	Same forecasts; 21 comparisons

The complete episode used 12 model calls, with reported cost $1.243572. There were no automatic retries or adoptions. The numerical comparator searched 21 source/summary combinations, versus Lab's allowance of six; equal scores here do not establish matched-compute equivalence. Actions varied: the unchanged control tested B04 participation; the candidate tested B04 and B06 concentration. All forecasts remained zero. The receipt reached the next request exactly, but the reviewer repeated its unsupported proposal. Neither action variation nor successful delivery demonstrated a useful repair.

This motivated a separately versioned engineering correction: code supplies ordered numerical evidence, the active feedback task is explicit, and its contract admits continuation decisions rather than a fresh initial proposal. An action signature must also be executable: commitment has no source-basket field. One further call ($0.375084) withdrew the numerical diagnosis and rejected the proposal, but still treated the advisory minimum as mandatory and misstated the candidate's commit step. The original evidence remains preserved. This engineering correction does not establish reliable review or autonomous learning.

Scope and next evidence. The earlier 25/28/3-node experiment and 512-character edits remain historical comparators. The next substantive result must connect a supported diagnosis to a changed investigation and protected-case performance.

February, May and August packet cases are exposed development material. Baskets share market conditions; overlapping horizons create dependence; historical replay cannot exclude model pretraining knowledge. The dataset lacks certified historical release and adjustment vintages. These limits preclude trading-performance claims.

DSPy composes language-model programmes [2]; MIPRO optimises their instructions and demonstrations [3]; TextGrad propagates textual feedback through graphs [4]. These established methods frame Zenith's question about located diagnoses and useful repairs. Source-bound execution evidence is in `docs/investigator-evidence.json` and `docs/status.md`.

References

[1] SocrateAI. *NEPSE Open Data*, pinned source `497e27238a10`. Dataset-specific MIT-form licence.

[2] O. Khattab et al. *DSPy*. 2023. arXiv:2310.03714.

[3] K. Opsahl-Ong et al. *Optimizing Instructions and Demonstrations for Multi-Stage Language Model Programs*. 2024. arXiv:2406.11695.

[4] M. Yuksekgonul et al. *TextGrad: Automatic Differentiation via Text*. 2024. arXiv:2406.07496.